

Generation Interconnection System Impact Study Report

For Queue Project Q20-1596

Newport Tie 230 kV

March 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1596. The Interconnection Customer proposes a 319.1 MW solar photovoltaic facility interconnecting at the Newport Tie 230 kV substation in the Ohio area. The request was received on October 12, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 510 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$57,620,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Anaverde – Vtulli 100kV	Canal Rd – Newport Tie 230kV	132.7	141	9.3
Himes – Anaverde 100kV	Canal Road – Newport Tie 230kV line	126.1	138	9.3
Madisonburg – Newport Tie 100kV Line 1	Newport Tie – Madisonburg 100kV Cir. 2	118.8	161	5.7
Newport Tie – Madisonburg 100kV Circuit 2	Newport Tie – Madisonburg 100kV Line 1	118.8	161	5.7
Newport Tie – Madisonburg 100kV	Madisonburg – Newport Tie 100kV Circuit 1	118.8	161	5.7
Newport Tie 230 kV/100 kV Autotransformer	Newport Tie 230/100kV Transformer Bank	106.5	229	8.3
Newport Tie 230 kV/100 kV Autotransformer Bank 2	Newport Tie 230/100 kV Auto T1	106.5	229	8.3
Newport Tie 230 kV/100 kV Autotransformer	Newport Tie 230-100 kV Transformer Bank 1	106.5	229	8.3
Himes – Nampa 100kV	Newport Tie – Cana Rd 230kV line	104.6	128	9.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vitulli – Anaverde 100kV	Rebuild line with higher-capacity conductor	\$2,229,000	100% (\$2,229,000)
Himes – Anaverde 100kV	Rebuild line with higher-capacity conductor	\$4,088,000	100% (\$4,088,000)
Madisonburg – Newport Tie 100kV Line 1	Rebuild line with higher-capacity conductor	\$6,267,000	100% (\$6,267,000)
Newport Tie – Madisonburg 100kV line	Rebuild line with higher-capacity conductor	\$6,924,000	100% (\$6,924,000)
Newport Tie – Madisonburg 100kV Ckt 3	Rebuild line with higher-capacity conductor	\$6,969,000	100% (\$6,969,000)
Newport Tie 230 kV/100 kV Autotransformer	Replace transformer with higher-rated unit	\$9,349,000	100% (\$9,349,000)
Newport Tie 230 kV/100 kV Autotransformer Bank 2	Replace transformer with higher-rated unit	\$12,479,000	100% (\$12,479,000)
Newport Tie 230/100kV TX #3	Replace transformer with higher-rated unit	\$8,223,000	100% (\$8,223,000)
Himes – Nampa 100kV	Reconductor line	\$1,092,000	100% (\$1,092,000)
	Total allocated to project		\$57,620,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.